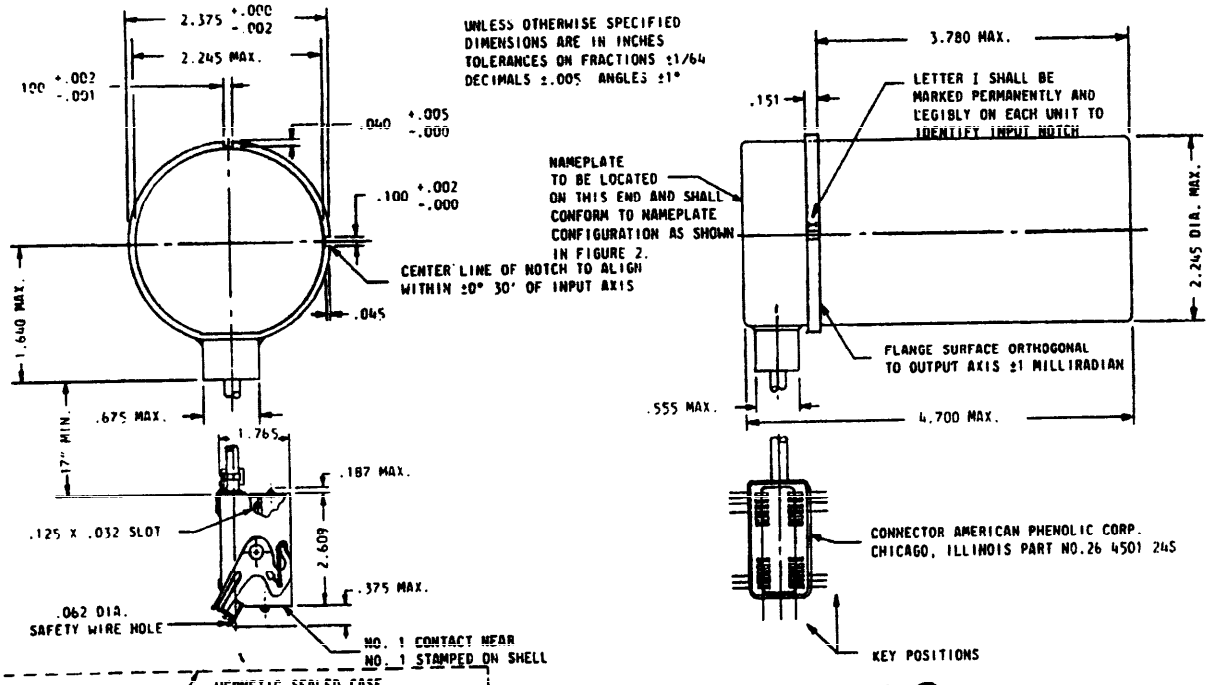
MIL-G-81168/1A(OS)
19 April 1967
SUPERSEDING
MIL-G-81168/1(WP)
5 February 1965
BUORD DWG 617432 (A)

MILITARY SPECIFICATION SHEET
GYROSCOPE, RATE INTEGRATING

THE COMPLETE REQUIREMENTS FOR PROCURING THE GYRO DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-81168. IN THE EVENT OF CONFLICT BETWEEN THIS DOCUMENT AND ANY OF THE DOCUMENTS FORMING A PART OF THIS SPECIFICATION, THE REQUIREMENTS OF THIS SPECIFICATION SHALL GOVERN.



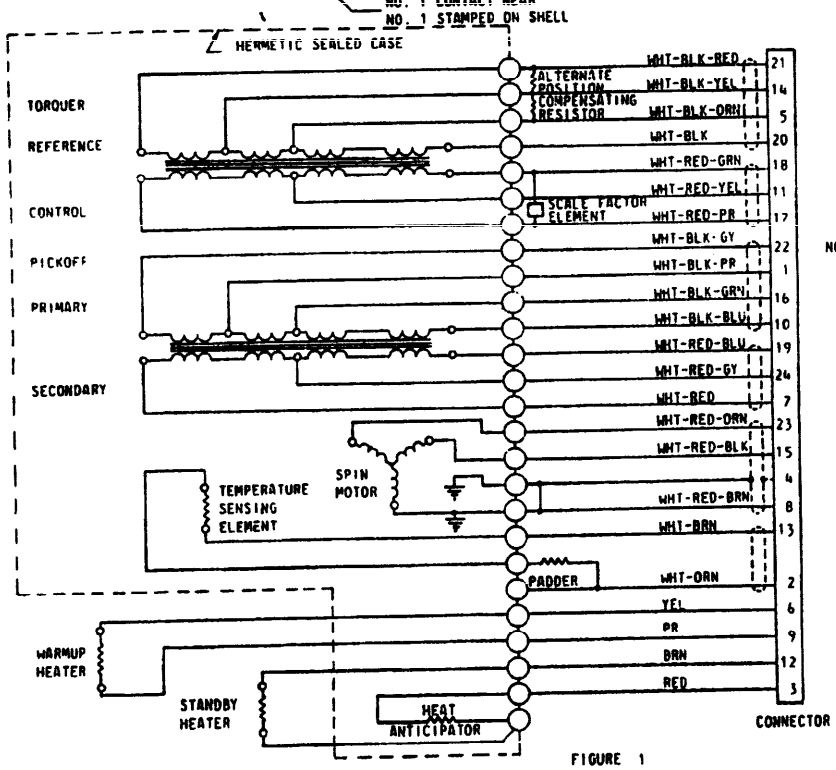


FIGURE 1

NOTES:

1. IT IS RECOMMENDED THAT CONNECTOR PINS 7, 10, 18 AND 21 BE GROUNDED FOR MINIMUM CAPACITANCE EFFECTS AT GYRO NULL.
2. CABLE SHIELDS SHALL BE INSULATED FROM THE CONNECTOR HOOD.
3. COLOR CODING OF LEAD WIRES: PRIMARY COLOR - BASE STRIPE - NARROW STRIPE.
4. THE OUTLINE CONFIGURATION IS FOR REFERENCE ONLY.
5. THE GYRO SHALL BE WARRANTED FOR 2000 HOURS OF SPIN MOTOR OPERATION AT OPERATING TEMPERATURE, OR FOR ONE YEAR, WHICHEVER OCCURS FIRST.
6. THE GYRO SHALL HAVE A MINIMUM SHELF LIFE OF 3 YEARS AT AMBIENT TEMPERATURES WITHIN THE SPECIFIED LIMITS.
7. TEST MOUNTING ACCURACY: AXIS ALIGNMENT TESTS ± 0.5 ARC MINUTES. ALL OTHER TESTS ± 0.5 DEGREES.

(A) For changes see pages 1 and 4.

REVISED

FSC-5840 (A)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

Dielectric Strength: 500 volts rms, 60 cps at the following test points:

7 to case	6 to case
10 to case	2 to case
7 to 10	6 to 2
18 to case	3 to case
21 to case	2 to 3
18 to 21	

Insulation Resistance: Terminal test points same as for Dielectric Strength. Test potential 500 volts d.c.

Spin Motor			
Type:	synchronous, hysteresis		
Winding Configuration:	3 phase		
Rated Excitation:			
start	10.5	volts	±1.0
run	9.0	volts	±1.5
	Spin motor shall run at synchronous speed from 380 to 420 cps.		
Test Excitation:*	10.0	volts	±0.5
	400	cps	±0.1%
	2	% distortion	maximum
Angular Velocity:	12,000	rpm	nominal
Rotor Angular Momentum	2.0×10^4	$gm\text{-}cm^2/sec$	nominal
Run-up Time	15	sec	maximum
Power Consumption:	3.0	watts	maximum
	0.5	power factor	±0.1

Pickoff			
Type:	four pole, variable reluctance microsyn		
Rated Excitation:	0-100	ma	range
Test Excitation:*	100	ma	±0.5%
	400	cps	±0.1%
	2	% distortion	maximum
Impedance:			
primary	9.9 + j5.0	ohms	±12%
	with open circuit secondary		
secondary	2130 + j4520	ohms	±12%
	with open circuit primary		
Null Voltage:	10	mv rms	maximum
Phase Shift:	67	degrees	±5
Scale Factor:	26.5	mv/mr	nominal
Linearity Error	5	%	maximum
	measured at input angles of 0, ±1, ±2, and ±3 degrees.		

For changes see paras 1 and 4

(A) *Note: Spinmotor, pickoff, torquer and heater excitations shall be supplied from the same frequency source.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

Torquer

Type:	four pole, variable reluctance microsyn		
Reference Winding Excitation:	0-170	ma	range
Control Winding Excitation:	0-145	ma	range
Test Excitation:*	100	ma	±1%
	400	cps	±0.1%
	2	% distortion	maximum
Command Rate Scale Factor:	0.1000	mr/sec/ma^2	±0.5%

based on ± inputs to 5000 ma^2 inclusive.

Torquer Linearity Error:

±Input ma^2	Deviation mr/sec	Linearity Error percent
250	.45	.020
750	.85	.038
2000	1.85	.082
5000	4.25	.189
10000	12.25	.544

Maximum Command Rate:	2250	mr/sec	nominal
Impedance:			
reference winding	130 + j282	ohms	±12%
	with open circuit control winding		
control winding	188 + j302	ohms	±12%
	with open circuit reference winding		

Temperature Sensor

Type:	resistance alloy wire		
Scale Factor:	1.5	ohms/°F	±15%
Operating Temperature Resistance:	780	ohms	±1
Induced Voltage:	3.0	mv rms	maximum
Test Excitation:	10	ma	maximum
	2	% distortion	maximum
Operating Temperature:	165	°F	±5

Heaters

Power:			
Warm-up heater	75	watts	minimum
Standby heater	101	watts	±12%
Test Excitation:*	115	volts	±1%
	400	cps	±0.1%

Gyro Unit

Gyro Scale Factor:	25	mv/mr	±10%
Input Angular Freedom:	6.5	deg	minimum
	from pickoff null		

Gyro Phasing:

Spinmotor — With an ABC phase sequence, positive spinmotor rotation will result with terminal 15 connected to line A, terminal 23 connected to line B, and terminal 8 connected to line C.

Ⓐ For changes see pages 1 and 4"

Source: https://assist.dla.mil -- Downloaded: 20260-23T05:01Z
Check the source to verify that this is the current version before use.

Pickoff:	Positive output voltage exists when voltage from terminal 7 to terminal 19 leads the voltage from 10 to terminal 22 by less than 90 degrees.		
Torquer:	Positive torque is obtained when terminals 18 and 21 are grounded and the currents to terminals 17 and 20 are in phase.		
Characteristic Time:	3.0	ms	maximum
Axis Alignment:	OA orthogonal to mounting flange surface within 1 milliradian. IA aligned within ±0.5 degree maximum of the IRA notch when pickoff is at null.		
Acceleration Insensitive Drift:	60	deg/hr	maximum
Acceleration Sensitive Drift: test procedures	45 4.27 of MIL-G-81168	deg/hr	max. resultant
Day to Day Stability:	20	deg/hr	maximum
Ambient Temperature:	-40 to +165	degrees F	range
Operating Life:	2000	hrs	minimum

Vibration, shock, altitude and moisture resistance shall be conducted in accordance with MIL-G-81168.

Nameplate: Gyros shall be identified with markings conforming to MIL-STD-130. Markings shall be applied by acid, electric etching, permanent marking ink or engraving applied directly to the case. The markings shall be as permanent as the normal life expectancy of the gyro on which it is applied and legibility shall be as required for readability and identification. Markings shall withstand the same environmental tests required of the gyro.

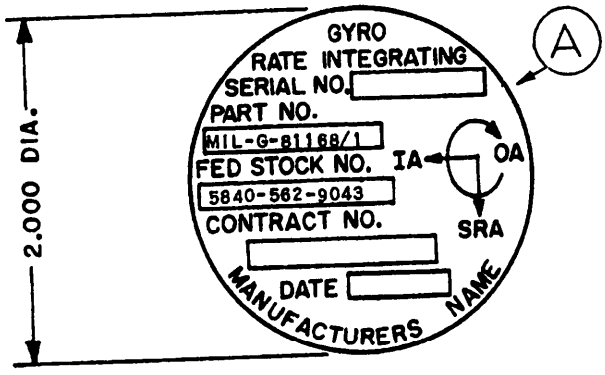


FIGURE 2

NAMEPLATE CONFIGURATION

Ⓐ For changes see pages 1 and 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-G-81168/1A(OS) GYROSCOPE, RATE INTEGRATING

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 83 1426

0-8879

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL ORDNANCE SYSTEMS COMMAND (ORD 9343)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.